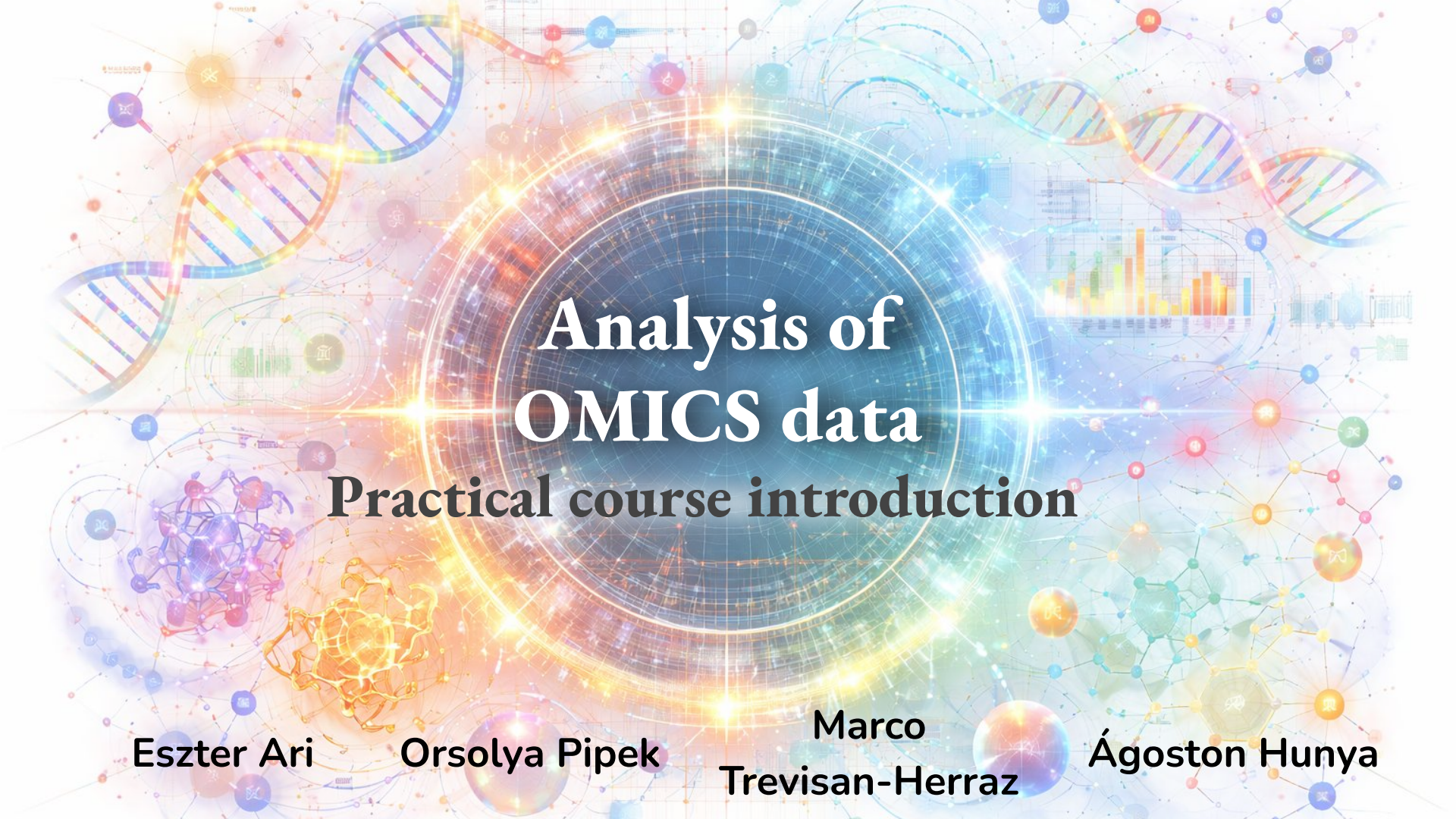




Analysis of OMICS data

Practical course introduction

Eszter Ari

Orsolya Pipek

**Marco
Trevisan-Herraz**

Ágoston Hunya

Meet your instructors

Eszter Ari

ELTE, Department of Genetics

 arieszter@ttk.elte.hu

Topics: NGS, Transcriptomics, Functional enrichment



Orsolya Pipek

ELTE, Department of Physics of Complex Systems

 orsolya.pipek@ttk.elte.hu

Topics: Bash/Linux, Genomics

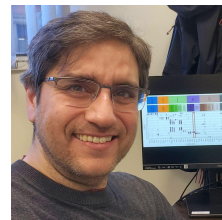


Marco Trevisan-Herraz

Newcastle University

 marco.trevisan@newcastle.ac.uk

Topics: Proteomics, ChIP-seq, Single-cell RNAseq

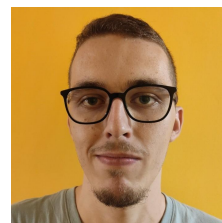


Ágoston Hunya

ELTE, Department of Genetics

 ago.hunya@gmail.com

Topics: Metagenomics (16S rRNA)



What is OMICS?

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The **collective** and **comprehensive** characterization and **quantification** of **entire sets** of biological molecules.

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examining all the proteins in a cell or organism, the **proteome**
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etc, etc, etc...

What is OMICS?



MASSIVE
datasets

What is OMICS?



**MASSIVE
datasets**



**computational
analysis**

What is OMICS?



**MASSIVE
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**computational
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↶ This is what
you'll learn!

Requirements

Weekly grades (50%)

tutorials, **smaller
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during courses

(finish during class or
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comprehensive,
well-documented analysis of a chosen
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+ **presentation** of the
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(a minimum grade
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- MSc: max. 3
- PhD: max. 6

→ no grade → ignored in mean

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Submitting late:

→ 1-grade
deduction weekly

Project work timeline



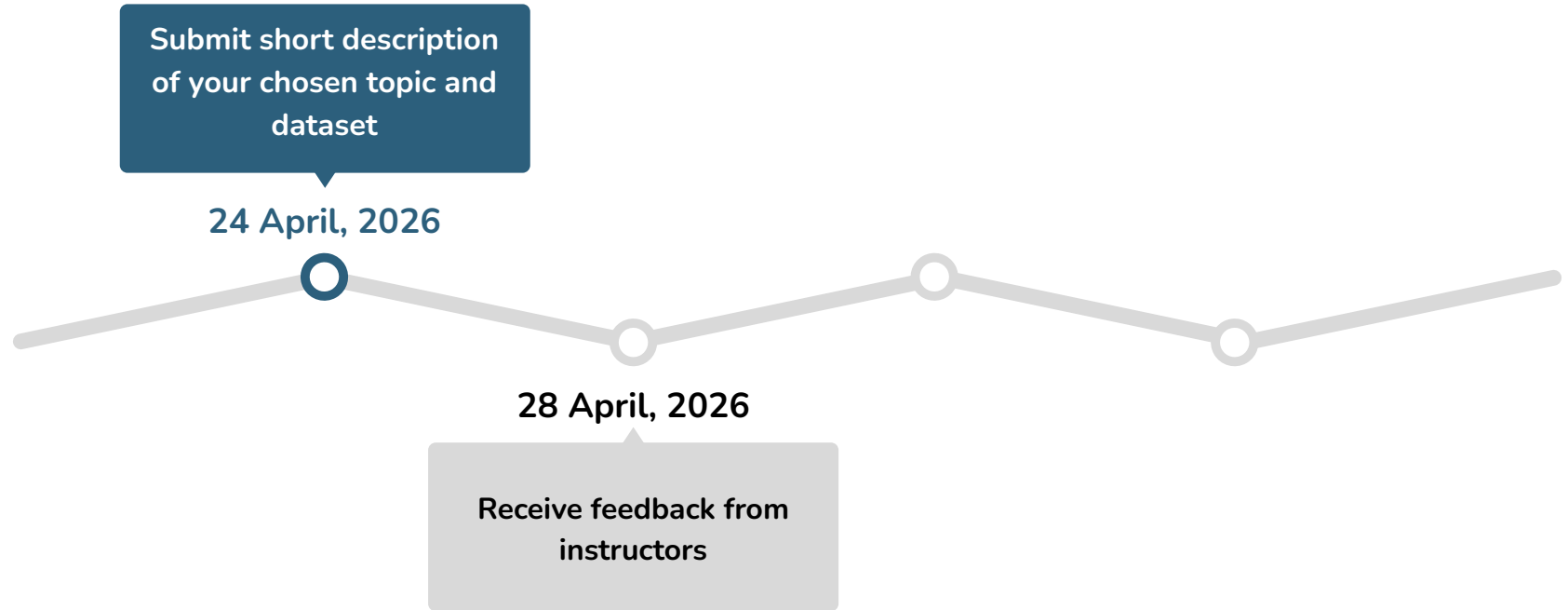
Project work timeline

Submit short description
of your chosen topic and
dataset

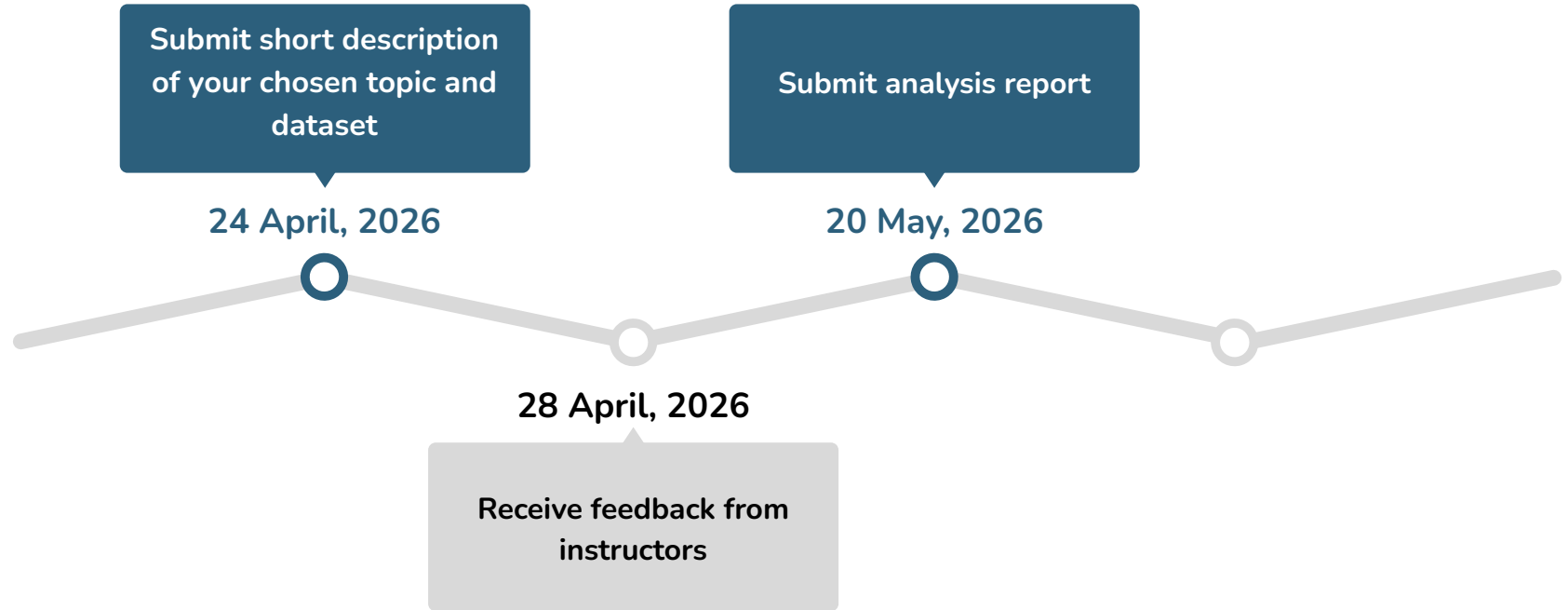
24 April, 2026



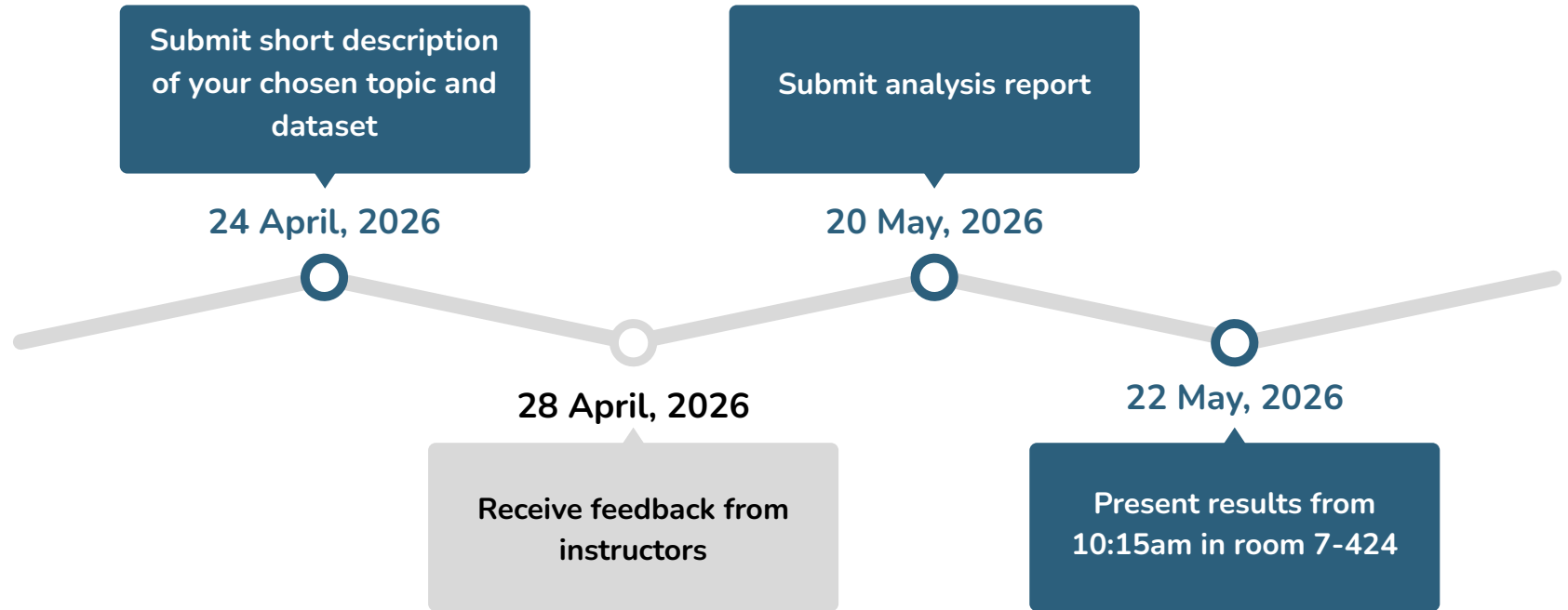
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Project work timeline



Resources & Tips

Useful links:

-  Online attendance:
<https://teams.microsoft.com/meet/33091260655366?p=xoByG1mo1VoQcD0xcp>
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Be patient and don't panic



Make mistakes, that's how you learn!



Ask AI to explain things you don't understand (it's great!)



Ask for help



Always document your work



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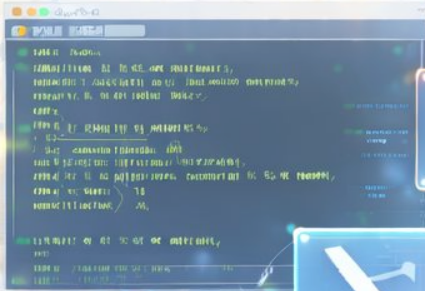
- Ask instructors
- **Help each other**
- Use MS Teams chat

Outside practicals:

- MS Teams discussions
- Email instructors
- Office hours (by appointment)



Why Linux & command line?



The challenge of OMICS data

The scale of modern biology

The scale of modern biology



Traditional biology

- Small datasets
- Excel spreadsheets
- Point-and-click software
- Visual inspection
- Single computer

Example:

analyzing 100 genes

File size: ~100 KB

Time: minutes

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- **MASSIVE** datasets
- Command-line tools
- Automated pipelines
- Computational analysis
- Server/cluster computing

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whole genome sequencing

File size: 50-200 GB

Time: hours to days

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Limitations of GUI



Power of command line

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2. **Not reproducible** (hard to document exact steps)
3. **Not automatable** (must repeat manually)
4. **Memory intensive** (crash with large files)
5. **Not transferable** (different on each OS version)



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3. **Automatable** (run overnight, run 1000x)
4. **Efficient** (minimal memory usage)
5. **Universal** (works on multiple systems)

When clicking fails

Task: Find all genes on chr1 with expression > 100 in 50 samples



Limitations of GUI



Power of command line

When clicking fails

Task: Find all genes on chr1 with expression > 100 in 50 samples



Limitations of GUI

1. Open Sample1.txt in Excel
2. Scroll to chromosome 1
3. Filter by expression > 100
4. Copy results → new file
5. Repeat for Sample2.txt
6. Repeat for Sample3.txt
- ...
50. Repeat for Sample50.txt
51. Merge all files manually



Time: ~3-4 hours

Reproducible: ✗ **Error-prone:** ✓



Power of command line

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Power of command line

```
for sample in Sample*.txt; do
  grep '^chr1' $sample | \
    awk '$3 > 100'
done | cut -f2 | sort -u \
  > chr1_genes.txt
```



Time: 2 seconds

Reproducible: ✓ **Error-prone:** ✗

When clicking fails

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5000 samples? No problem!

Time: 2 seconds

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What is a server?



RAM	8-16 GB	256+ GB
storage	0.5-1 TB	10+ TB
CPU cores	2-8	64+
GPU	basic	high-performance (A100/V100)
handling TBs of data	✗	✓
fast parallel processing	✗	✓
runs 24/7	✗	✓

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secure shell
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server address
(~ which building to look for)

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secure shell command

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Goals for today

What you'll learn

- Connect to the dlab server
- Navigate the Linux file system
- Create and organize directories
- Manipulate files (copy, move, delete)
- Search and filter data
- Download large datasets
- Use Jupyter notebooks

Useful advice

- Type, don't copy-paste (builds muscle memory!)
- **Read and understand** error messages
- Use tab completion (saves time, prevents typos)
- Google, ChatGPT, Claude, Gemini, etc. are your friends
- Ask questions early

Let's get started!

➤ Step 1

Open tutorial

➤ Step 2

Connect to server

➤ Step 3

Work through tasks

➤ Step 4

Complete and submit Jupyter notebook within 1 week

Get the
tutorial!

https://pipeko.web.elte.hu/teaching/OMICS/2026_OMICS01_bash.html

Find your
username!

https://docs.google.com/spreadsheets/d/1M3BDOqr6tmxso59QehTbgTYVG2AdO_lLgCrEkzcYGlVU/edit?usp=sharing

